

Machine Learning Based Road Crash Severity Prediction and Risk Pattern Analysis Using Victorian Crash Data

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING BACHELOR OF TECHNOLOGY

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Abstract

This project develops and evaluates a machine learning system for predicting road crash severity using Victorian road crash data. The study uses a large tabular dataset containing crash, road, vehicle, road-user, location and temporal attributes. The workflow covers data inspection, leakage-aware preprocessing, exploratory data analysis, information-gain-based feature selection, class-imbalance experiments, supervised classification, hyperparameter tuning, K-Means crash archetype discovery, and feature-interaction experiments. Multiple linear and ensemble models are compared using metrics that are appropriate for the imbalanced three-class target.

Keywords: road crash severity, machine learning, class imbalance, feature selection, ensemble learning, explainable AI

1. Introduction

1.1 Purpose of the Project

Road crashes produce outcomes ranging from other injuries to serious injuries and fatalities, and the severity of a crash depends on interacting factors such as crash type, speed environment, road geometry, road-user involvement, location and time. For transport agencies, researchers and road-safety analysts, a severity-prediction system can help identify patterns associated with higher-risk crash outcomes and support evidence-based analysis of safety data.

The main motivation for this project is the difficulty of learning severity from highly imbalanced crash data. In the project dataset, Fatal crashes represent only about 1.67% of usable records, while Other Injury and Serious Injury account for most observations. A model can therefore achieve reasonable raw accuracy while performing poorly on the minority class. The project consequently evaluates macro-averaged precision, recall and F1-score in addition to accuracy and multiclass ROC-AUC.

1.2 Problem Statement

The problem is formulated as a three-class supervised classification task. Given information available in the crash record, the system predicts one of three severity categories: Fatal, Other Injury, or Serious Injury. Four non-injury records were removed because the class contained only four observations in the working dataset. The expected output for a new crash record is a predicted severity class together with class probabilities were supported by the selected model.

1.3 Existing Methods and Disadvantages

Conventional statistical and simple machine-learning models can be easier to interpret but may struggle with nonlinear relationships, high-dimensional categorical variables and interactions among crash factors. The Victorian crash study supplied with the project reported that an initial Decision Tree achieved about 61% accuracy while the confusion matrix and F1-score showed strong majority-class behavior. Recent studies also report that crash-severity datasets are commonly imbalanced, especially for fatal outcomes, which can reduce minority-class sensitivity.

Modern ensemble models can capture nonlinear relationships but introduce their own challenges: they may overfit, become difficult to interpret, and still favor the majority class if imbalance is not considered. High accuracies reported in other datasets are not directly comparable when their class distributions and data sources differ substantially. For example, one supplied ensemble paper reports about 92.26% accuracy, but its Fatal class forms roughly 61–64% of the dataset, whereas Fatal is only 1.67% in this project.

1.4 Proposed System

The proposed system uses a leakage-aware preprocessing pipeline followed by information-gain feature selection and a sequence of supervised experiments. Logistic Regression is used as the baseline, followed by class-weighted and oversampled Logistic Regression, Random Forest, and advanced ensembles including AdaBoost, Gradient Boosting, HistGradientBoosting, XGBoost, LightGBM and CatBoost. Random Forest tuning uses stratified three-fold RandomizedSearchCV with macro F1 as the optimization metric. K-Means++ clustering is additionally used to discover crash archetypes and test whether cluster membership improves severity prediction. The current ensemble comparison identifies HistGradientBoosting as the strongest tested model on the existing 136-feature representation, while feature-interaction experiments are being evaluated before final model selection.

1.5 Research Objectives

1. Prepare and validate a suitable Victorian road-crash dataset for three-class severity prediction.
2. Perform systematic preprocessing and exploratory analysis while documenting missingness, duplicates and class imbalance.
3. Prevent target leakage and select informative predictors using training-only Information Gain.
4. Develop baseline, imbalance-aware, tree-based and advanced ensemble severity classifiers.
5. Tune selected models using stratified cross-validation and macro F1 as the optimization objective.
6. Evaluate models using accuracy, macro precision, macro recall, macro F1, ROC-AUC and confusion matrices.
7. Discover crash archetypes using K-Means++ and test whether cluster membership improves prediction.
8. Investigate feature interactions and explain model behavior before finalizing the deployable model.

S.No.	Title	Reference with DOI	Problem or gap addressed	Objective of the Research	Focus of the paper (Methods used)	Inference	Recommendations for future research
1	Ensemble machine learning for traffic accident injury severity prediction with explainable AI	2026 medRxiv preprint. DOI: 10.64898/2026.04.13.26350778	Need for accurate and interpretable multi-class injury-severity prediction using ensemble learning.	Compare ensemble classifiers and interpret predictions with SHAP.	AdaBoost, XGBoost, LightGBM, HistGBRT, CatBoost, Gradient Boosting, NGBoost and Random Forest; class-wise metrics and SHAP.	Reported ensemble accuracies were around 92.2%; HistGBRT reported 92.26% in that dataset; SHAP supported global and local interpretation.	Test robustness on datasets with substantially different class distributions and stronger imbalance-aware protocols.
2	Machine learning approaches to traffic accident severity prediction: Addressing class imbalance	Amiri, Afshari & Soltani (2025), Machine Learning with Applications, 22, 100792. DOI: 10.1016/j.mlwa.2025.100792	Rare fatal pedestrian outcomes create severe imbalance.	Compare data-, algorithm-, and decision-level imbalance handling for Australian pedestrian severity prediction.	Logistic Regression, SVM, Decision Tree, XGBoost; SMOTE, ADASYN, ROS, class weighting, threshold moving; SHAP.	Threshold moving with XGBoost achieved 72% fatality F1 and 84% macro-accuracy in the study setting.	Validate balancing methods across other Australian crash datasets and severity definitions.
3	AI-based prediction of traffic crash severity for improving road safety and transportation efficiency	Mostafa et al. (2025), Scientific Reports, 15, 27468. DOI: 10.1038/s41598-025-10970-7	Large crash datasets contain heterogeneous human, vehicle and crash factors and class imbalance.	Build a scalable ML framework integrating feature engineering, clustering, imbalance handling	Feature engineering; K-Means/HDBSCAN; RandomOverSampler/SMOTE variants; CFS/RFE; Extra Trees and other classifiers.	Extra Trees reached 96.19% accuracy and 95.28% macro-F1 in the reported dataset after the proposed processing pipeline.	Test robustness on external datasets and investigate deployment/real-time use.

S.No.	Title	Reference with DOI	Problem or gap addressed	Objective of the Research	Focus of the paper (Methods used)	Inference	Recommendations for future research
				and feature selection.			
4	Predicting Traffic Crash Severity in Jeddah Using Machine Learning Techniques: A Data-Driven Approach to Road Safety	Alanazi & Okail (2025), Transportation Research Record, 2679(9), 324–343. DOI: 10.1177/03611981251337460	Urban crash severity depends on interacting driver, road, weather and violation factors.	Compare multiple ML models for RTA severity using 877 crashes and 22 independent variables.	Decision Tree, Random Forest, LightGBM, XGBoost, CatBoost and AdaBoost.	Demonstrates comparative use of modern tree ensembles for crash severity in an urban Saudi setting.	Use larger multi-year datasets and stronger imbalance-aware validation.
5	A methodological framework for road accident severity prediction for Indian highways using machine learning models	Khanum, Garg & Faheem (2025), MethodsX, 15, 103728. DOI: 10.1016/j.mex.2025.103728	Traditional models may miss nonlinear relationships and provide limited interpretability.	Develop an interpretable RF/GB framework for Indian highway accident severity.	Random Forest, Gradient Boosting, SHAP feature selection/explanation; class-wise evaluation and confusion analysis.	RF achieved 86.65% test accuracy; SHAP identified vehicle, location and road-condition factors as influential.	Evaluate on more highway corridors and integrate richer environmental/traffic variables.
6	Nonlinear effects of traffic statuses and road geometries on highway traffic accident severity: A machine learning approach	Liang, Yuan, Wang, Wan, Liu, Wu, Chen & Tang (2024), PLOS ONE, 19(11), e0314133. DOI: 10.1371/journal.pone.0314133	Linear statistical assumptions may not capture nonlinear and interaction effects of traffic and road geometry.	Identify nonlinear, threshold and interaction effects related to crash severity.	RF, GBDT, XGBoost, LightGBM; partial dependence plots for single and paired factors.	Truck share, traffic volume, slope length, road grade, curvature and curve length showed important nonlinear/threshold effects.	Examine interaction effects across more road classes and geographic settings.
7	Comparison of road crash severity prediction using tree-based machine learning algorithms with varying data treatments	Acharya & Henry (2024), Intelligence, Informatics and Infrastructure, 5(1), 15–24. DOI: 10.11532/jsceii.5.1_15	Crash severity is difficult to predict because many crash, vehicle and road factors interact; imbalance can cause minority underprediction.	Compare Decision Tree and Random Forest under two vehicle-feature treatments.	Decision Tree and Random Forest; comparison of vehicle-type-only versus vehicle-type-plus-count data treatments.	Models showed stronger prediction of majority classes and difficulty with minority/fatal cases, illustrating the impact of imbalance.	Test resampling and cost-sensitive methods and include richer crash-context variables.
8	Comparison of traffic accident injury severity prediction models with explainable machine learning	Cicek, Akin, Uysal & Topcu Aytas (2023), Transportation Letters, 15(9), 1043–1054. DOI: 10.1080/19427867.2023.2214758	Complex ML models require interpretation of feature effects.	Compare injury-severity models and identify important contributing factors.	Decision Tree, MLP, SVC, Case-Based Reasoning, Naive Bayes; SHAP-based explanation.	Speed violations, alcohol consumption and belt usage were among influential factors; MLP had the highest accuracy in that study.	Use larger datasets and compare explanation stability across models.
9	Evaluating the	Obasi & Benson (2023), Heliyon, 9(8), e18812. DOI:	Traditional statistical	Compare several	Naive Bayes, Random Forest,	RF and Logistic Regression	Extend the study using casualty-

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	effectiveness of machine learning techniques in forecasting the severity of traffic accidents	10.1016/j.heliyon.2023.e18812	approaches can have limited predictive performance on complex crash data.	ML algorithms for UK accident-severity prediction .	Regression and ANN; RF feature importance; accuracy, precision, recall and F1.	achieved 87% overall accuracy on the reported UK dataset.	level information and richer injury variables.
10	Evaluating the effectiveness of machine learning techniques in forecasting the severity of traffic accidents	Obasi & Benson (2023), Heliyon, 9(8), e18812. DOI: 10.1016/j.heliyon.2023.e18812	Traditional statistical approaches can have limited predictive performance on complex crash data.	Compare several ML algorithms for UK accident-severity prediction .	Naive Bayes, Random Forest, Logistic Regression and ANN; RF feature importance; accuracy, precision, recall and F1.	RF and Logistic Regression achieved 87% overall accuracy on the reported UK dataset.	Extend the study using casualty-level information and richer injury variables.

3. Methodology

3.1 Proposed System Flow Diagram

Proposed System Flow for Road Crash Severity Prediction

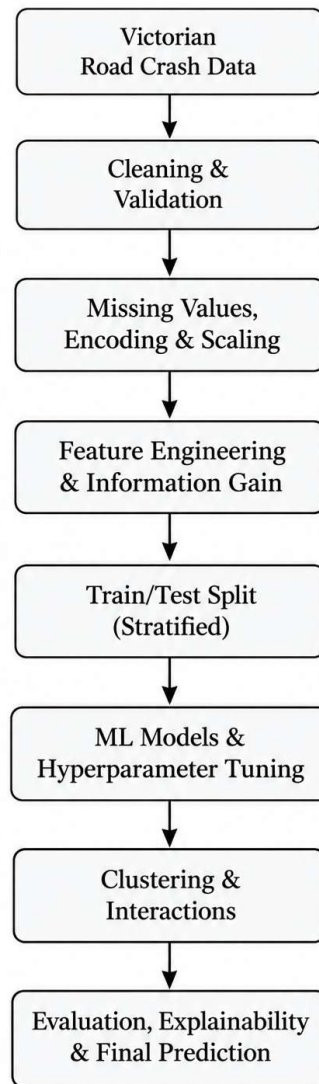


Figure 1. End-to-end project workflow: dataset preparation → leakage-aware preprocessing → feature selection → model training/tuning → evaluation → clustering/interaction analysis → explainability and final prediction.

3.2 Dataset Description

The working dataset is the Victorian Road Crash Data extract published by the Victorian Department of Transport and Planning. The official dataset contains crash information covering time, location, conditions, crash type and road-user attributes. The local CSV used in this project contains 200,352 records and 52 columns, with crash dates from 1 January 2012 through 31 December 2025. The target is SEVERITY. After removing four Non-Injury records and outcome-derived leakage columns, the modelling matrix contains 200,348 usable records. The target distribution is Fatal: 3,352 (1.673%), Other Injury: 124,942 (62.361%), and Serious Injury: 72,054 (35.964%). The raw data contains numerical, categorical, date/time and identifier fields.

3.3 Experimental Setup

Experiments were developed as Jupyter notebooks in a Windows 11 environment using Python 3.13.7. The development machine used an AMD Ryzen 5 7000-series CPU, 24 GB RAM and an NVIDIA RTX 3050 6 GB GPU. Core libraries included pandas, NumPy, SciPy, scikit-learn, imbalanced-learn, Matplotlib, Seaborn, XGBoost, LightGBM and CatBoost; NGBoost was also tested. Sparse matrices were used where appropriate to control memory use. Random seeds were fixed at 42 for reproducibility in the principal experiments.

3.4 Data Preprocessing

- Data inspection and schema validation: 200,352 rows, 52 columns, mixed numerical/categorical/date/time fields.
- Duplicate analysis: no duplicate ACCIDENT_NO values were found.
- Missing-value treatment: numerical columns use training-set medians; categorical values use an explicit Missing category.
- Outlier treatment: no blanket outlier deletion was applied; extreme values were retained unless they were identified as invalid target records.
- Categorical encoding: 16 low-cardinality categorical features were one-hot encoded; ROAD_NAME was replaced with train-only frequency encoding.
- Numerical scaling: StandardScaler was fit only on the training data and then applied to the test data.
- Target preparation: four non-injury rows were removed because the class contained only four observations; remaining labels were encoded as Fatal=0, Other Injury=1, Serious Injury=2.
- Leakage prevention: injury-derived outcome columns were removed before model fitting; train-dependent transformations were fit only on X_train.

- **Train/test split: stratified 80/20 split with random_state=42, producing 160,278 training and 40,070 test records.**

EDA examined data types, missingness, duplicate identifiers, descriptive statistics, class distribution, numerical distributions, categorical distributions and correlations. The target is strongly imbalanced, with Fatal crashes forming only 1.673% of usable observations. Missingness is concentrated in SRNS (about 70%), RMA/DIVIDED (about 3.83%), and ROAD_TYPE (about 1.37%), with smaller amounts elsewhere. No duplicate ACCIDENT_NO values were found. Correlation analysis was interpreted cautiously because injury-derived columns can create direct target leakage. The EDA therefore informed feature selection and interaction candidates rather than being used as the sole basis for selecting predictors.

3.6 Feature Engineering and Feature Selection

Date and time fields were transformed into YEAR, MONTH, DAY, DAY_OF_WEEK_NUM and HOUR. ROAD_NAME was converted to a train-only frequency feature (ROAD_NAME_FREQ). Information Gain was then computed using only the training data. One-hot encoded features were grouped back to their original feature names so that selection was performed at the original-feature level. A practical threshold of $IG \geq 0.005$ retained 26 original features, which corresponded to 136 encoded columns.

The selected original features include variables such as POLICE_ATTEND, DCA_CODE, DRIVER, LONGITUDE, ROAD_ROUTE_1, PASSENGERVEHICLE, NO_OF_VEHICLES, VICGRID_Y, LATITUDE, MOTORCYCLIST, DCA_CODE_DESCRIPTION, ACCIDENT_TYPE, PEDESTRIAN, YOUNG_DRIVER_18_25, RUN_OFFROAD, DEG_URBAN_NAME, MOTORCYCLE, SPEED_ZONE, OLD_DRIVER_75_AND_OVER, TOTAL_PERSONS, OLD_PED_65_AND_OVER, FEMALES, BICYCLIST, MALES and ROAD_NAME_FREQ. Outcome-derived injury-count variables were excluded before modelling to prevent leakage.

3.7 Model Building and Training

The baseline model is multinomial Logistic Regression using the 136 selected encoded features. It is followed by class-weighted and RandomOverSampler experiments to examine imbalance handling. Random Forest is evaluated with bootstrap aggregation, random feature subsampling and out-of-bag scoring. Advanced ensemble experiments include AdaBoost, Gradient Boosting, HistGradientBoosting, XGBoost, LightGBM and CatBoost.

The current HistGradientBoosting configuration uses max_iter=75, learning_rate=0.05, max_leaf_nodes=31 and l2_regularization=1.0. The XGBoost configuration uses 75 estimators, depth 6, learning rate 0.05, subsample 0.8, colsample_bytree 0.8 and histogram tree construction. LightGBM uses 75 estimators, learning rate 0.05 and 31 leaves. CatBoost uses 75 iterations, depth 6 and learning rate 0.05.

3.8 Hyperparameter Tuning

Random Forest hyperparameter tuning used RandomizedSearchCV with StratifiedKFold(n_splits=3, shuffle=True, random_state=42), 10 sampled parameter combinations, macro F1 scoring and n_jobs=1. The selected configuration was n_estimators=300, max_features=0.5, max_depth=None, min_samples_split=5 and min_samples_leaf=1. The best cross-validation macro F1 was approximately 0.4140. This tuned model achieved 0.6516 accuracy, 0.4915 macro precision, 0.4124 macro recall, 0.4146 macro F1 and 0.7432 multiclass ROC-AUC on the held-out test set.

3.9 Model Evaluation

Evaluation uses accuracy, macro precision, macro recall, macro F1-score, multiclass one-vs-rest ROC-AUC, classification reports and confusion matrices. Macro averaging is emphasized because each severity class is given equal weight, which prevents the large Other Injury class from dominating the summary metric. Confusion matrices are used to examine which severity classes are confused, with particular attention to Fatal recall.

4. Results and Discussion

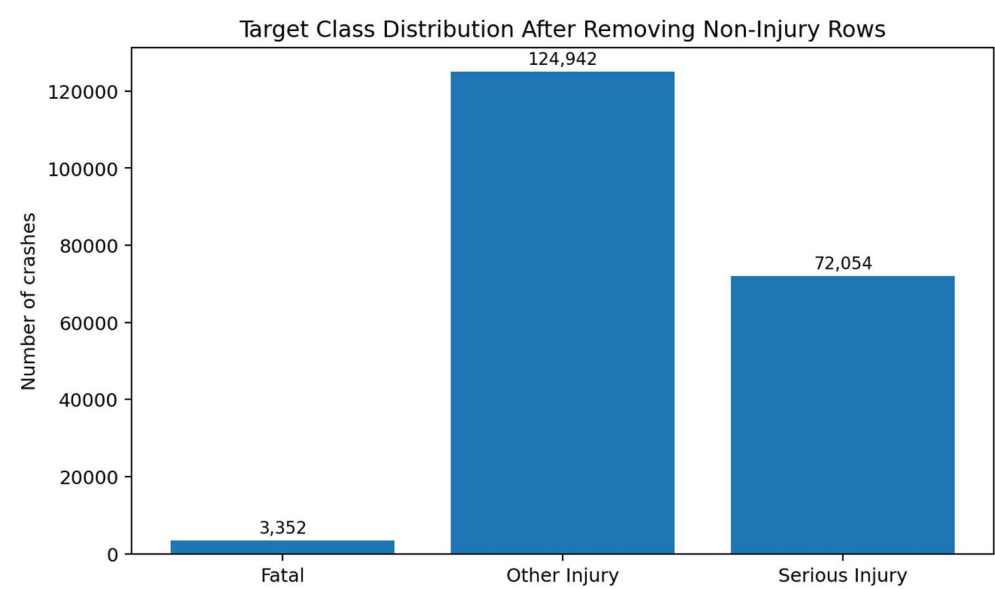
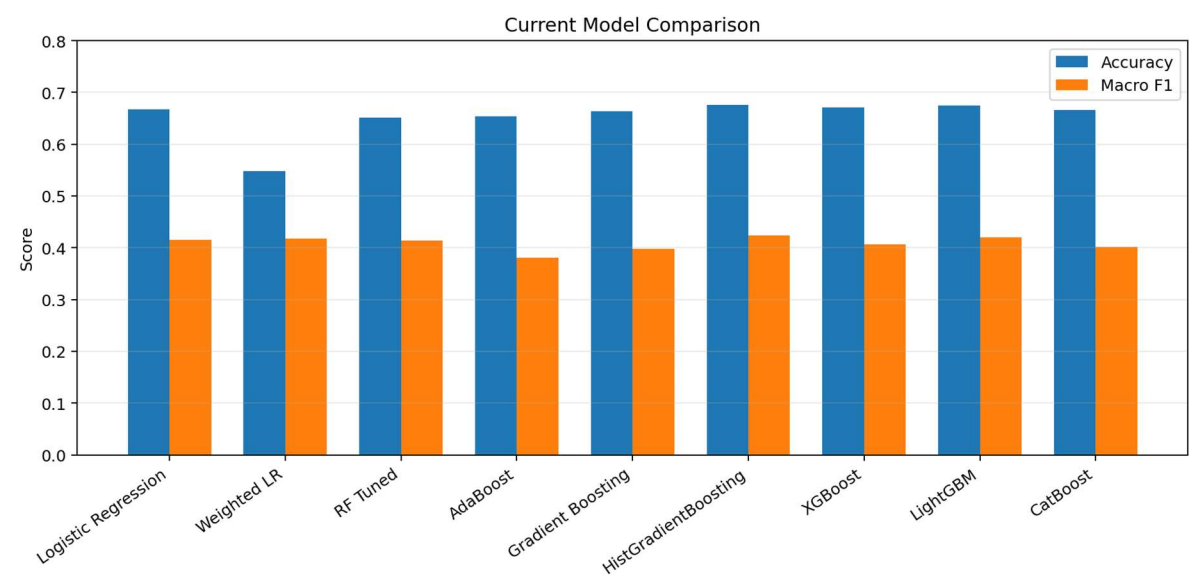
4.1 Preprocessing Results

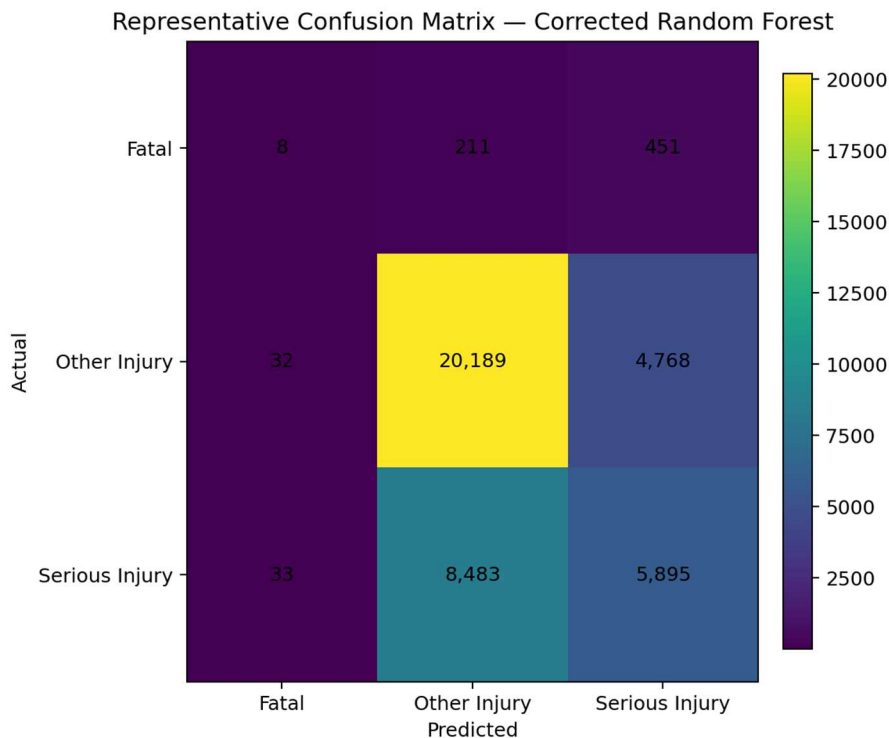
The raw dataset contains 200,352 rows and 52 columns. Four non-injury records were removed, leaving 200,348 records for modelling. Five direct outcome-derived columns were removed from predictors, together with SEVERITY as the target and ACCIDENT_NO as an identifier. The data was split before train-dependent transformations: 160,278 training records and 40,070 test records using a stratified 80/20 split. Training-only median imputation, categorical Missing-value handling, ROAD_NAME frequency encoding, one-hot encoding and StandardScaler fitting were applied. The final pre-IG encoded representation had 361 features; information gain retained 26 original features represented by 136 encoded columns. The final train/test matrices contain no NaN or infinite values.

4.2 Model Performance

Model	Accuracy	Precision	Recall	F1-score	ROC AUC
Existing study — HistGBRT (2026 preprint)	92.26%	—	—	—	—
Logistic Regression baseline	66.74%	0.5271	0.4146	0.4150	0.7664
Weighted Logistic Regression	54.85%	0.4409	0.5874	0.4184	0.7336
Random Forest tuned	65.16%	0.4915	0.4124	0.4146	0.7432
AdaBoost	65.35%	0.4102	0.3892	0.3803	0.7135
Gradient Boosting	66.34%	0.6195	0.4022	0.3982	0.7603
HistGradientBoosting	67.58%	0.6791	0.4232	0.4245	0.7759
XGBoost	67.09%	0.4257	0.4099	0.4065	0.7711
LightGBM	67.49%	0.7618	0.4200	0.4205	0.7742

Model	Accuracy	Precision	Recall	F1-score	ROC AUC
CatBoost	66.66%	0.4221	0.4055	0.4013	0.7629





The representative Random Forest confusion matrix shows the minority-class limitation: only 8 of 670 Fatal test cases were correctly classified in this model, while many Fatal cases were predicted as Other Injury or Serious Injury. This motivates the continued focus on macro metrics, class-specific recall and interaction/imbalance analysis.

5. Conclusion

This project established a complete leakage-aware machine-learning pipeline for three-class Victorian road crash severity prediction. The workflow covers data cleaning, missing-value treatment, temporal feature extraction, categorical encoding, scaling, information-gain feature selection, imbalance experiments, Random Forest tuning, advanced ensemble comparison and K-Means crash archetype analysis. On the current feature representation, HistGradientBoosting provides the strongest tested held-out result with 67.58% accuracy and 0.4245 macro F1. The analysis also demonstrates why accuracy alone is insufficient for this dataset: Fatal crashes are only 1.67% of usable records and remain difficult to identify. Final model selection should therefore be completed after the interaction-feature and explainability stages.

6. Future Work

- 1. Complete and validate the screened feature-interaction experiments and retain only interactions that produce meaningful held-out improvements.
- 2. Apply SHAP/global and local explainability to the final selected ensemble model.
- 3. Investigate threshold adjustment and cost-sensitive learning specifically for the rare Fatal class.
- 4. Perform deeper hyperparameter optimization using cross-validation without touching the final test set.
- 5. Validate the final pipeline on a temporally separated or external Victorian crash dataset.
- 6. Investigate model drift as newer monthly crash data become available.
- 7. Develop a deployable application or API for entering a new crash record and returning predicted severity and probabilities.

7. References

[Use one consistent style, preferably APA. Include every cited work and verify DOI and bibliographic information.]

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Signature of the Guide

Course Instructor

Dataset source: <https://discover.data.vic.gov.au/dataset/victoria-road-crash-data>